Citations: 1494

Learning RoI Transformer for Oriented Object Detection in Aerial Images

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IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019

DOI: 10.1109/CVPR.2019.00296.

Introduction: Horizontal RoIs

- **Challenges:** Complex backgrounds, and arbitrary oriented objects.
- **Limitations:** Causing severe feature misalignment for densely packed objects.
- **Consequence:** A single HRoI often contains multiple objects, confusing classification and localization.



Figure 1. An aerial image applied Horizontal ROI

Introduction: HROI v.s. RROI

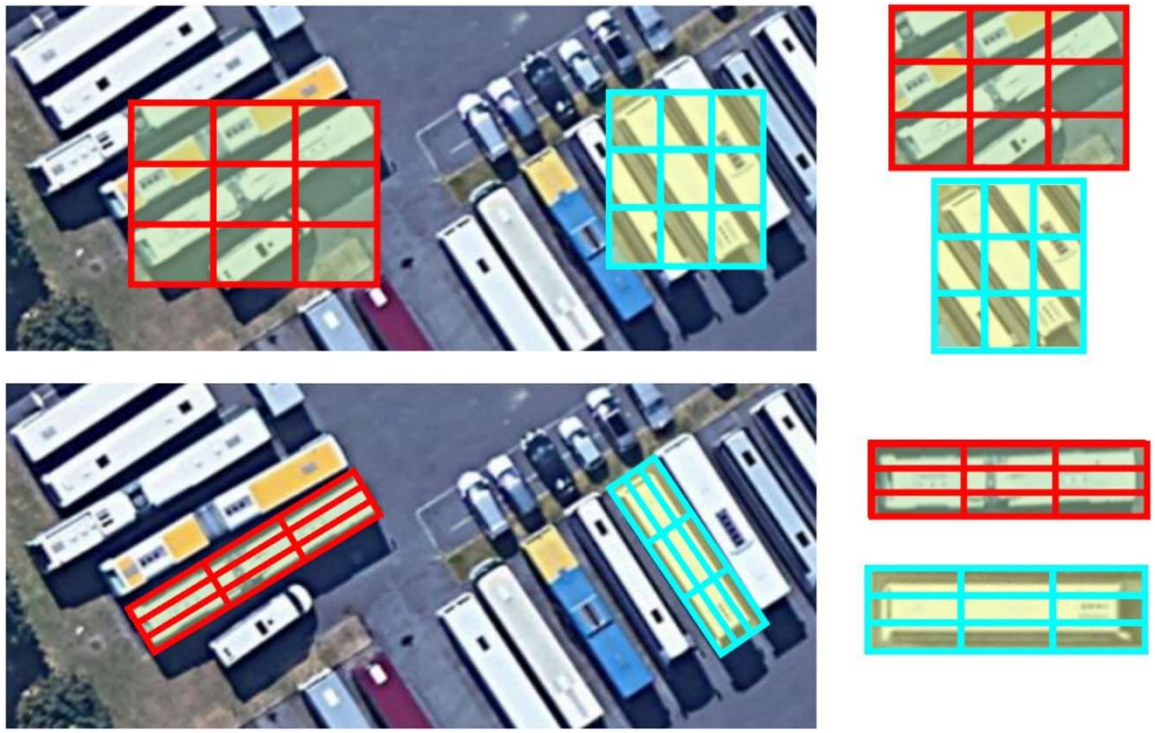


Figure 2. Horizontal (top) v.s. Rotated RoI warping (bottom)

Method: RoI Transformer

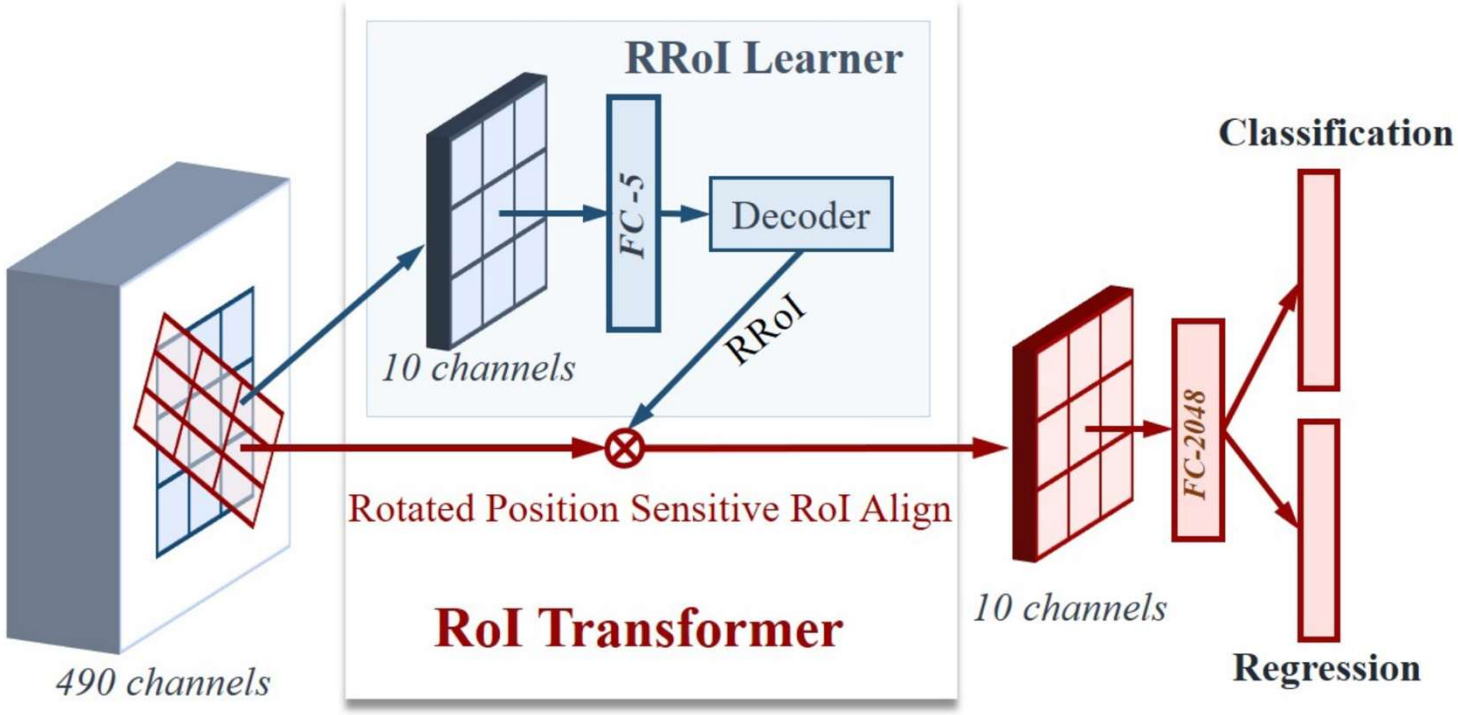


Figure 3. The architecture of RoI Transformer.

Method: Pipeline

Steps:

1. **Input:** Image patch of 1024 × 1024 × 3.
2. **Backbone:** ResNet-101 (optionally with FPN) produces the shared feature map.
3. **Light head:** Large separable conv (k = 15, C_mid = 256) gives the thin feature D: H × W × 490.
4. **RPN:** 15 horizontal anchors, 6000 proposals → NMS → 800 HRoIs (x, y, w, h).
5. **RRoI Learner:** PS RoI Align → FC(5) → box decoder → RRoI.
6. **RRoI Warping:** RPS RoI Align outputs Y: 7 × 7 × 10.
7. **Head:** One 2048-d FC, then sibling FCs for classification and OBB regression.

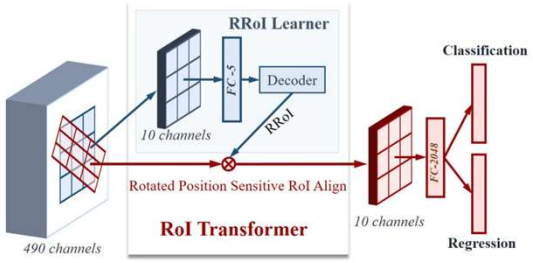


Figure 3. The architecture of RoI Transformer.

Method: Relative Offset

Problem: A global-coordinate offset gives a different number for the same relative position at every rotation angle.
Solution: Express the offset in the RRoI's own local frame

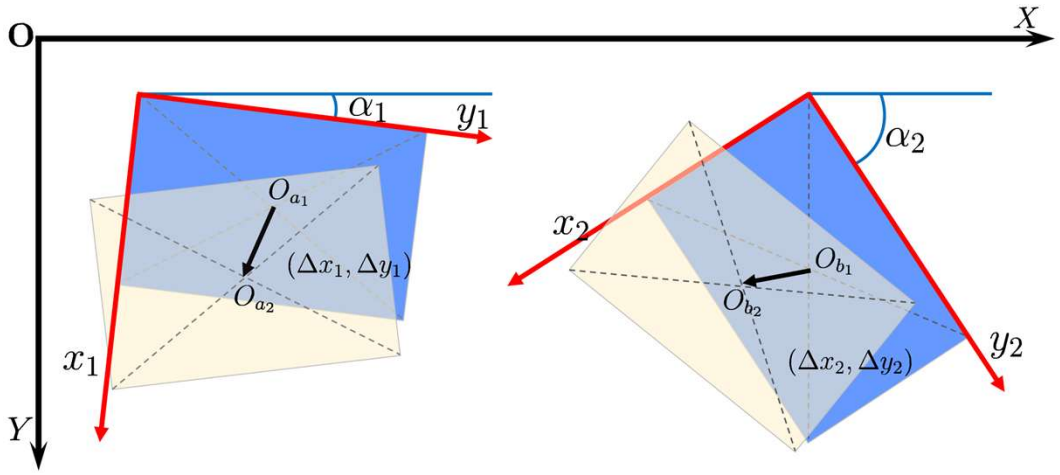


Figure 4. An example explaining the relative offset.

Method: RRoI warping

Process: Split the RRoI into K × K bins in its own local frame, then rotate each sampling point back onto the feature map.
Interpolation: Sample locations land on non-integer coordinates, so values come from bilinear interpolation — the whole step stays differentiable.
Position sensitive: Output cell (i,j) reads only the matching channel (i,j) out of the K × K × C feature, inherited from R-FCN.

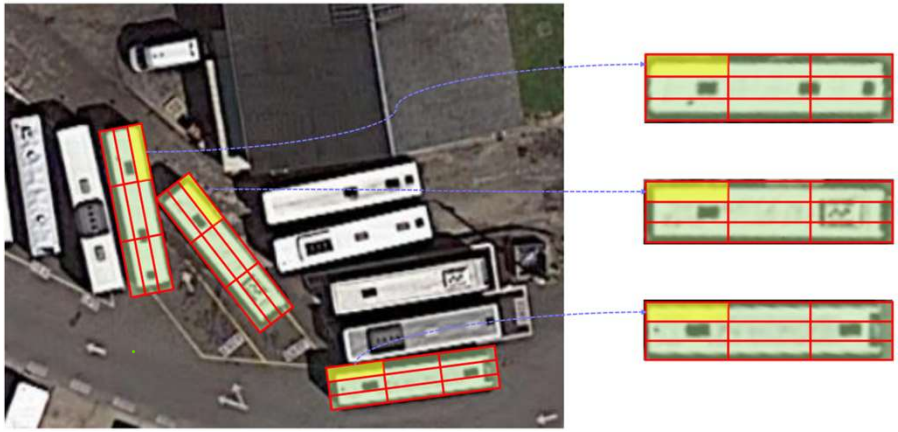


Figure 5. Rotated RoI warping

Datasets:

- **DOTA:**

The largest aerial image dataset, containing 2,806 large images, 188,282 instances, 15 categories.

- **HSRC 2016:**

Focused on ship detection, 1061 images and more than 20 categories of ships.

Datasets: Preprocessing

- **DOTA:**

1. Resized the images at scales 1.0 & 0.4 for training and scales 1.0 & 0.5 for testing.
2. Crop a series of 1024 × 1024 patches from the original images
3. Rotation augmentation randomly from 4 angles (0, 90, 180, 270)

- **HSRC 2016:**

1. Applied horizontal flipping
2. resized to (S, M) = (512, 800)

S = short side

M = maximum length

Results: HSRC2016

method	CP [26]	BL2 [26]	RC1 [26]	RC2 [26]	R^2PN [43]	RRD [20]	RoI Trans.
mAP	55.7	69.6	75.7	75.7	79.6	84.3	**86.2**

Table 1. Comparisons with the state-of-the-art methods on HRSC2016.

Results: DOTA

- **Baseline (Light-Head R-CNN OBB):** A fast two-stage detector modified to predict oriented boxes.
- **Competitor (DPSRP)**: Deformable PS RoI Pooling learns flexible spatial offsets to adapt geometric deformations.

Results: DOTA

method	backbone	W/FPN	test scales	SP	Ship	mAP
FR-O [37]	resnet101		1	47.4	37.16	54.13
RRPN [30]	resnet101		1	53.35	55.81	61.01
R2CNN [15]	resnet101		1	51.94	57.25	60.67
R-DFPN [41]	resnet101	✓	1	51.32	54.78	57.94
Yang et al. [42]	resnet101	✓	1	53.35	56.6	62.29
ICN [1]	dresnet101	✓	4	**64.17**	69.98	68.16
Baseline	resnet101		2	56.86	38.3	58.31
DPSRP	resnet101		2	64.04	53.56	63.89
RoITransformer	resnet101		2	61.29	66.57	67.74
Baseline	resnet101	✓	2	59.54	75.77	66.95
RoITransformer	resnet101	✓	2	58.93	**83.59**	**69.56**

Table 2. Comparisons with state-of-the-art detectors on DOTA

Results: DOTA

method	mAP	train speed	test speed	param
LR-O	58.3	**0.403 s**	**0.141s**	**273MB**
DPSRP	63.89	0.445s	0.206s	273.2MB
RT	**67.74**	0.475s	0.17s	**273MB**

Table 3. Comparison of our RoI Transformer with deformable PS RoI pooling and Light-Head R-CNN OBB on accuracy, speed and memory.

Results: DOTA

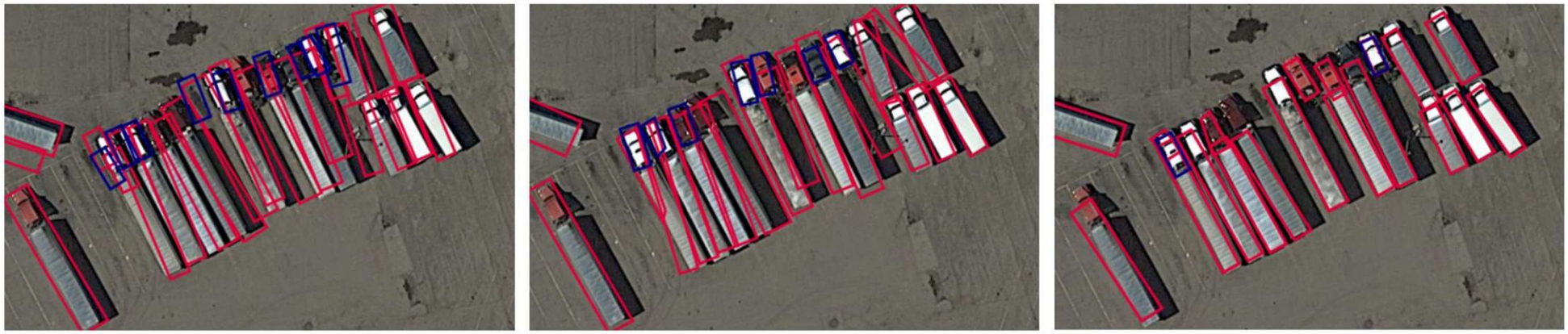


Figure 4. Visualization of detection on the scene where many densely packed instances exist.

Ablation Study on DOTA

- **Findings:**

1. Plugging in the module alone gains 4.87 mAP.
2. A precise RRoI keeps almost no background, so mild enlargement adds 2.86 mAP.
3. Skipping NMS on RRoIs keeps more proposals and raises recall, adding 1.5 mAP.

	Baseline	RoITransformer with different settings			
Light RRoI Learner?			✓	✓	✓
Context region enlarge?				✓	✓
NMS on RRoIS?		✓	✓	✓	
mAP	58.3	63.17	63.39	66.25	**67.74**

Table 4. Results of ablation studies. (Without FPN)

Conclusions

- **Application Limit:** Fails in ambiguous scenes without global context
- **Future Work**: Integrate broader contextual learning networks to distinguish objects from confusing backgrounds.

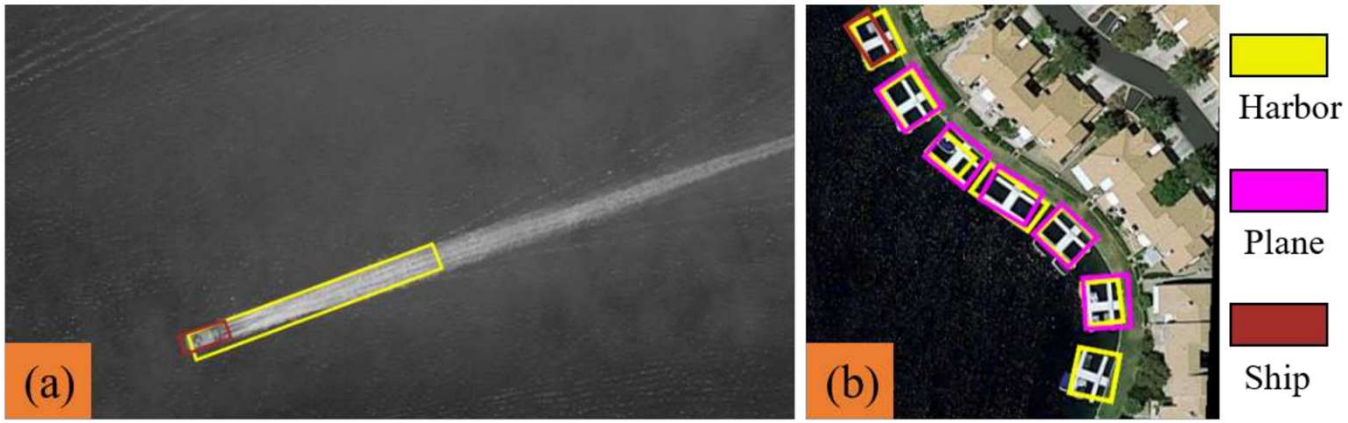


Figure 6. Failure cases.